

Yash Patil

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Education

Manipal University Jaipur

B.Tech, Computer Science Engineering (Data Science) – CGPA: 7.71/10

Class XII – 72% | Class X – 90%

Jaipur, India

Aug 2023 – Jun 2027

Publication

SA-PSO: Surrogate-Assisted Particle Swarm Optimization for Scalable Sentiment Feature Selection

Accepted at ADCIS 2026 (Springer LNNS), with Dr. Shitanshu Jain, Dept. of Data Science & Engineering, MUJ.

Skills

Analytics & Statistics:	Statistical Modeling, Predictive Analytics, Time-Series Forecasting, Monte Carlo Simulation, Risk Modeling (VaR, CVaR), Data Visualization
Optimization:	Particle Swarm Optimization, Surrogate-Assisted Optimization, Feature Selection
ML & GenAI:	Scikit-Learn, TensorFlow, Deep Learning, Reinforcement Learning, LLM Agents, Agentic Workflows, Prompt-Injection Defense
Languages & Data:	Python (NumPy, Pandas), SQL, C++, JavaScript, MongoDB
Engineering & Tools:	Node.js, Express.js, REST APIs, React.js, Next.js, Git, Linux, Postman, AI-assisted coding (Claude Code)

Experience

Jio Platforms Limited (JioCX)

Backend Developer Intern

Mumbai, India

May 2026 – Jul 2026

- Built scalable REST APIs and backend services powering customer-engagement and digital-support platforms.
- Redesigned MongoDB schemas and queries, improving query performance and application responsiveness.
- Implemented authentication, authorization, validation and centralized error handling; built reusable logging, monitoring and security middleware.
- Worked with cross-functional engineering teams in Agile sprints on production debugging, performance tuning and deployments.

Samarth Bharat Vyaaspeeth

Software Engineer

India

Dec 2025 – Jan 2026

- Designed Joi-based validation schemas and reusable middleware across APIs, forms and onboarding flows in Next.js/Node.js apps, strengthening data integrity and security.

Projects

Commodity Research Prediction System

[GitHub]

- Built an end-to-end commodity analytics pipeline for price forecasting and risk assessment on large-scale market data.
- Developed time-series predictive models; engineered volatility, trend and supply-demand indicators as features.
- Ran Monte Carlo simulations for scenario analysis and VaR/CVaR estimation; automated alerts for market-regime shifts.

Vanguard – Intent-Aware Security for Autonomous Financial Agents

[GitHub]

- Built a runtime authorization layer for AI agents: Intent Passports, spending policies and deterministic risk scoring with fail-closed ALLOW/REVIEW/BLOCK decisions.
- Defended against prompt injection, tool poisoning, duplicate payments, velocity anomalies and behavioral drift; quarantined agents on repeated suspicious activity.
- Integrated Razorpay (test mode) with HMAC-signed webhooks and idempotency; validated defenses via an Attack Battlebox test suite.

Tarang – Carbon Credit Verification Platform

[GitHub]

- Developed an AI-powered verification system that uses machine learning to assess the legitimacy of carbon-offset projects.
- Built blockchain smart-contract infrastructure in Solidity for carbon-credit tokenization and trading, improving transparency, traceability and trust across the transaction lifecycle.
- Designed the full-stack architecture using React, Node.js, MongoDB and Solidity.

Achievements & Leadership

- **Top Team** among 120+ shortlisted teams – Deloitte Capstone Project Challenge 2025.
- **2nd Place** – Avinya Eco-Tech Hackathon, IIT Guwahati; **Top 5** – HackX, Manipal University Jaipur.
- Co-Founder of a technology startup building AI-driven solutions.